

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed

Audience: Technology executives & network infrastructure leaders

Source: Cloudflare Radar — Year in Review 2024

Traffic growth and service popularity
Protocol and cryptography adoption shifts
Connectivity disruptions and resilience risks
Threat landscape and email security metrics

17% Global Traffic Growth and Starlink's 3.3x Surge

+17%

Global Internet traffic growth in 2024 (year-over-year)

3.3x

SpaceX Starlink traffic growth during 2024 (normalized)

Normalized Starlink traffic worldwide

Max

Notes: Trend line starts mid-January to normalize for post-holiday activity. Starlink scaled to peak observed request volume.



Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

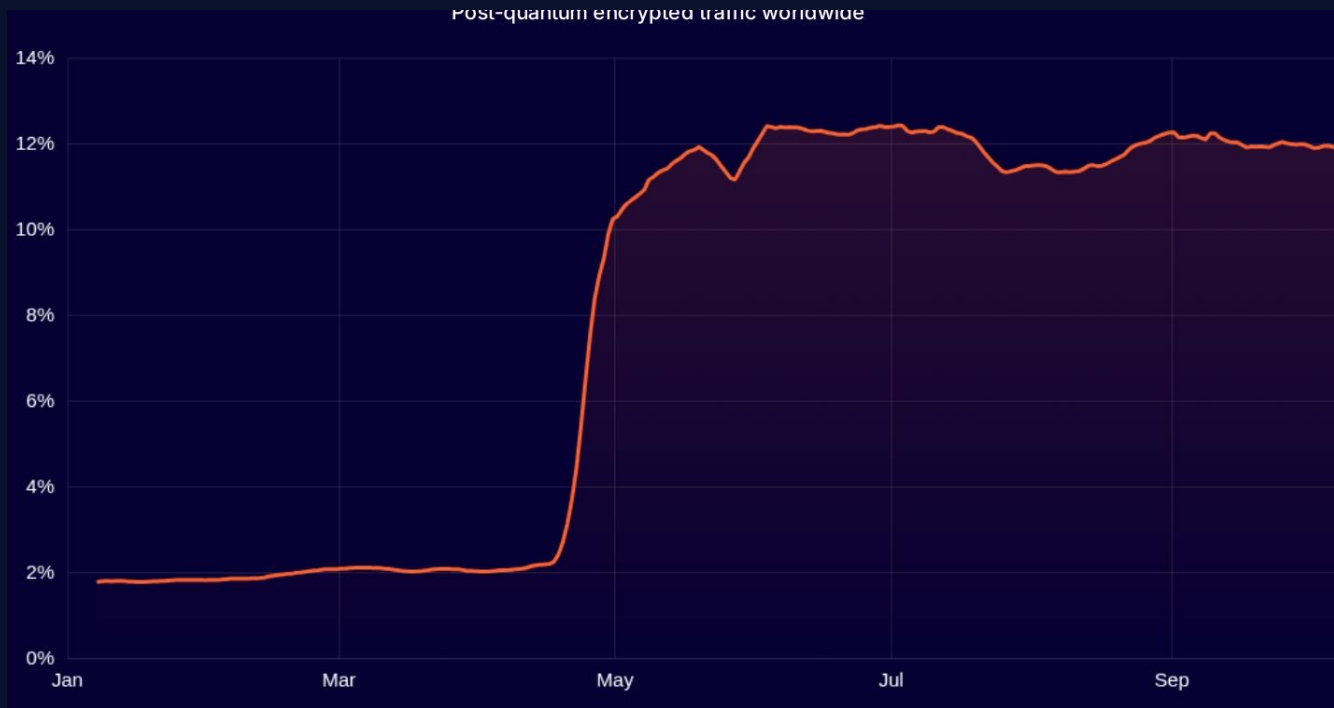
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Ranking based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

Source: The Cloudflare Radar 2024 Report: More New Top Services

Google remains #1 for the second consecutive year.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Post-quantum TLS 1.3

~2% → ~12–13%

Jump after Chrome 124 default enablement (Apr 17, 2024)

HTTP/3 adoption

21%

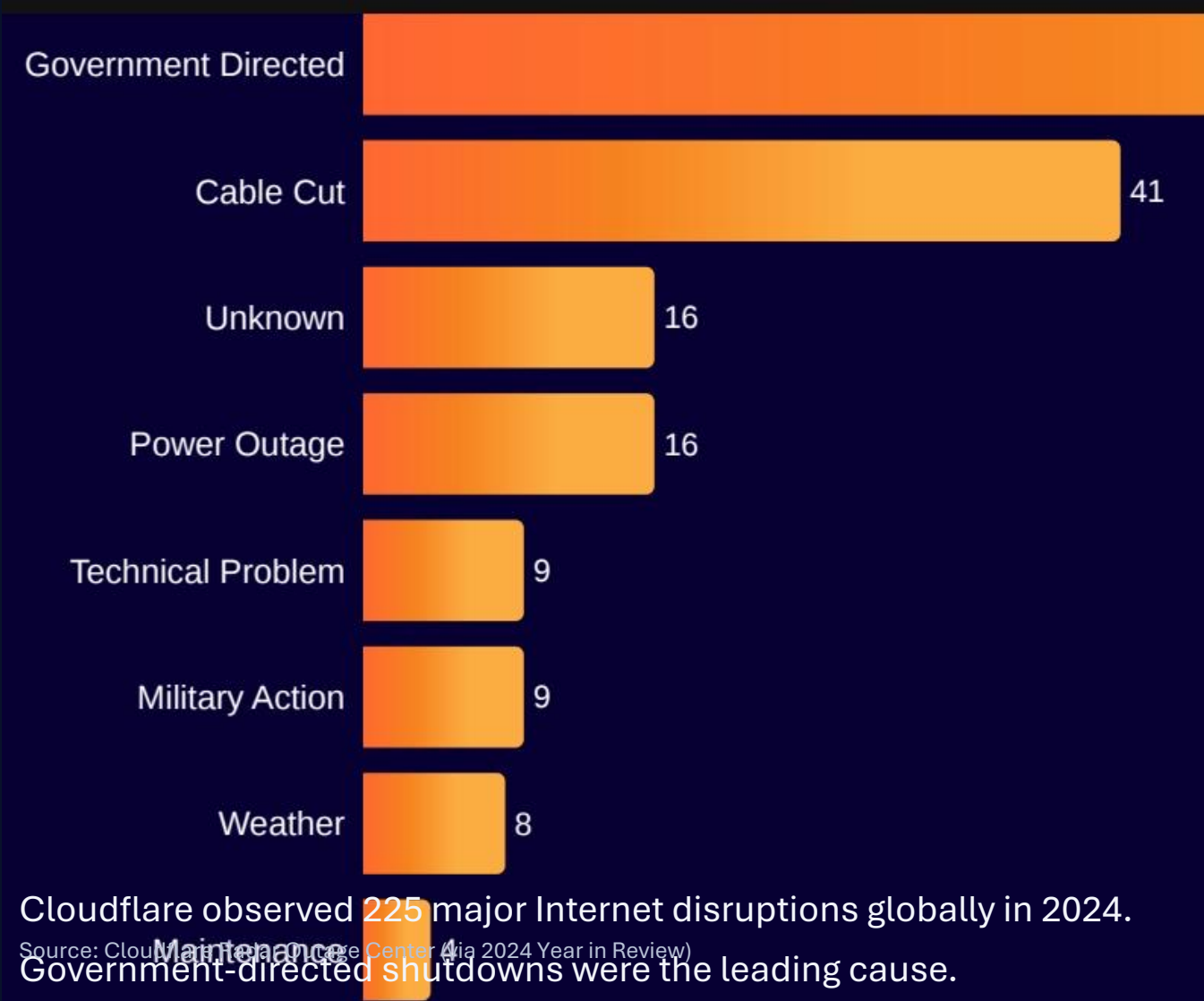
Share of traffic using HTTP/3 (QUIC)

Top API client language

Go (~12%)

Of automated API requests; Python 9.6%, Java 7.4%, .NET 3.6%

225 Major Disruptions: Government Shutdowns Lead Outage Causes



Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5% of traffic mitigated

Cloudflare mitigated 6.5% of all traffic it processed in 2024.
Most attacked sector: Gambling and Games.

Bot concentration & known vulns

68% of global bot traffic originated from the top 10 countries.
Log4j (CVE-2021-44228) exploitation persists years after disclosure.

Email security metrics (Cloudflare Email Security)

Metric	Value
Emails processed that were malicious	4.3%
Malicious emails containing a deceptive link	43%
Top-level domain with largest malicious/spam share	.bar

Five Takeaways Shaping the 2025 Internet Outlook

- Plan capacity for sustained traffic growth (+17% YoY) and rising satellite connectivity (Starlink 3.3×).
- Accelerate crypto agility: post-quantum TLS 1.3 moved from ~2% to ~13% after Chrome 124 (Apr 17, 2024).
- Harden for policy-driven disruption: government-directed shutdowns led the 225 major outages observed.
- Assume persistent exploitation: 6.5% of traffic was mitigated; Log4j attacks continue years later.
- Align platform strategy to the developer shift: Go leads API clients (~12%), while HTTP/3 reached 21% adoption.

2025 outlook: prioritize resilience, crypto agility, and modern protocol readiness.

Source: Cloudflare Radar 2024 Year in Review — synthesized takeaways